

Week 1 Wireshark to Investigate Network Traffic

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Introduction

For this assignment, I explored how to use ping, which is a command line networking utility which tests to see if a networked device is reachable. I sent requests to specific URLs and I got to know their IP addresses. I also learnt how to use Wireshark to capture packet data. This helped me analyze the traffic further and in detail ping could not provide. This was a form of enumeration as I got to find out more about the target URLs. ICMP is another protocol I have learnt about. It helps to communicate problems with data transmission in a network. Below, I show how I went about this while providing screen captures to support my answers.

Methodology

1. Does the source MAC address match your PC interface? Yes, it does
2. Does the destination MAC address in Wireshark match your team members MAC address? I did not get an IP address from a team member to ping so I opted to use Yahoo's URL instead. The Mac addresses are a match.
3. How is the MAC address of the pinged PC obtained by your PC?
IP addresses received for each URL?
IP address for www.yahoo.com: 188.125.88.206
MAC address for www.yahoo.com:
IP address for www.cisco.com: 2.17.169.15
MAC address for www.cisco.com:
IP address for www.google.com: 142.250.180.164
MAC address for www.google.com:

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Administrator: Command Prompt
C:\WINDOWS\system32>ping www.yahoo.com

Pinging me-ycpi-cf-www.g06.yahoodns.net [188.125.88.206] with 32 bytes of data:
Reply from 188.125.88.206: bytes=32 time=158ms TTL=54
Reply from 188.125.88.206: bytes=32 time=158ms TTL=54
Reply from 188.125.88.206: bytes=32 time=157ms TTL=54
Reply from 188.125.88.206: bytes=32 time=158ms TTL=54

Ping statistics for 188.125.88.206:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 157ms, Maximum = 158ms, Average = 157ms

C:\WINDOWS\system32>ping www.cisco.com

Pinging e2867.dsca.akamaiedge.net [2.17.169.15] with 32 bytes of data:
Reply from 2.17.169.15: bytes=32 time=17ms TTL=56
Reply from 2.17.169.15: bytes=32 time=15ms TTL=56
Reply from 2.17.169.15: bytes=32 time=13ms TTL=56
Reply from 2.17.169.15: bytes=32 time=15ms TTL=56

Ping statistics for 2.17.169.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 17ms, Average = 15ms

C:\WINDOWS\system32>ping www.google.com

Pinging www.google.com [142.250.180.164] with 32 bytes of data:
Reply from 142.250.180.164: bytes=32 time=140ms TTL=108
Reply from 142.250.180.164: bytes=32 time=140ms TTL=108
Reply from 142.250.180.164: bytes=32 time=140ms TTL=108
Reply from 142.250.180.164: bytes=32 time=140ms TTL=108

Ping statistics for 142.250.180.164:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 140ms, Maximum = 140ms, Average = 140ms

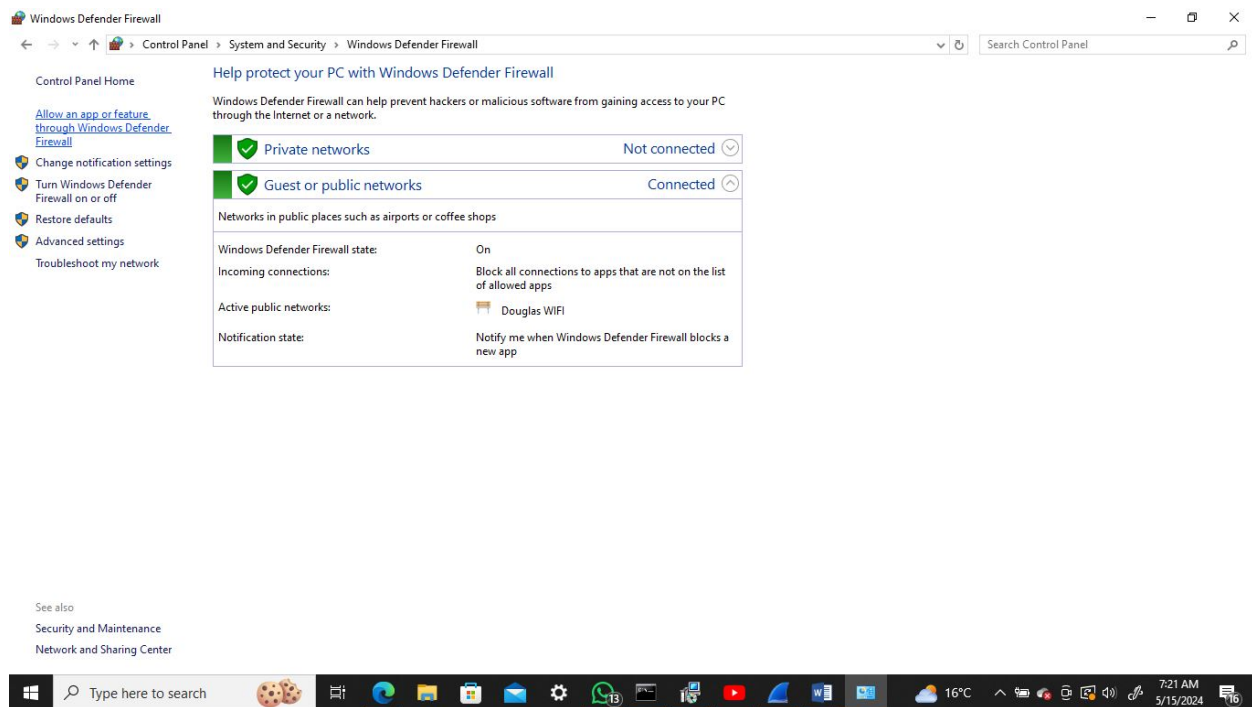
C:\WINDOWS\system32>
```

4. What is the significance of this information? IP and MAC addresses are both critical components of networking. They enable effective and efficient communication across and within networks. This ensures that data reaches its intended destination.
5. How does this information differ from the local ping information you received in part 1? It is more detailed than what I received in part 1.

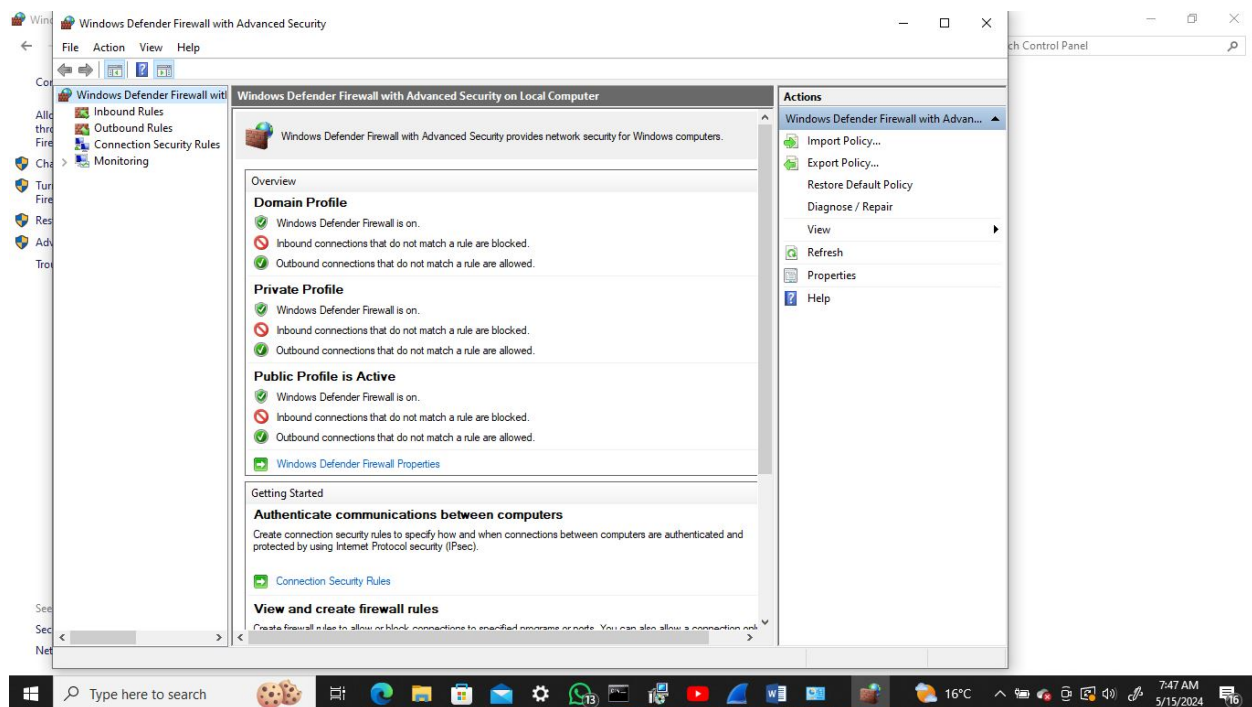
Reflection Question

Why does Wireshark show the actual MAC address of the local hosts, but not the actual Mac address for the remote hosts? This is because Wireshark captures frames at the Data Link Layer within the local network. For the remote hosts, Wireshark can only show MAC addresses for devices on the local network such as one's router and default gateway because MAC addresses do not travel beyond their local network segments.

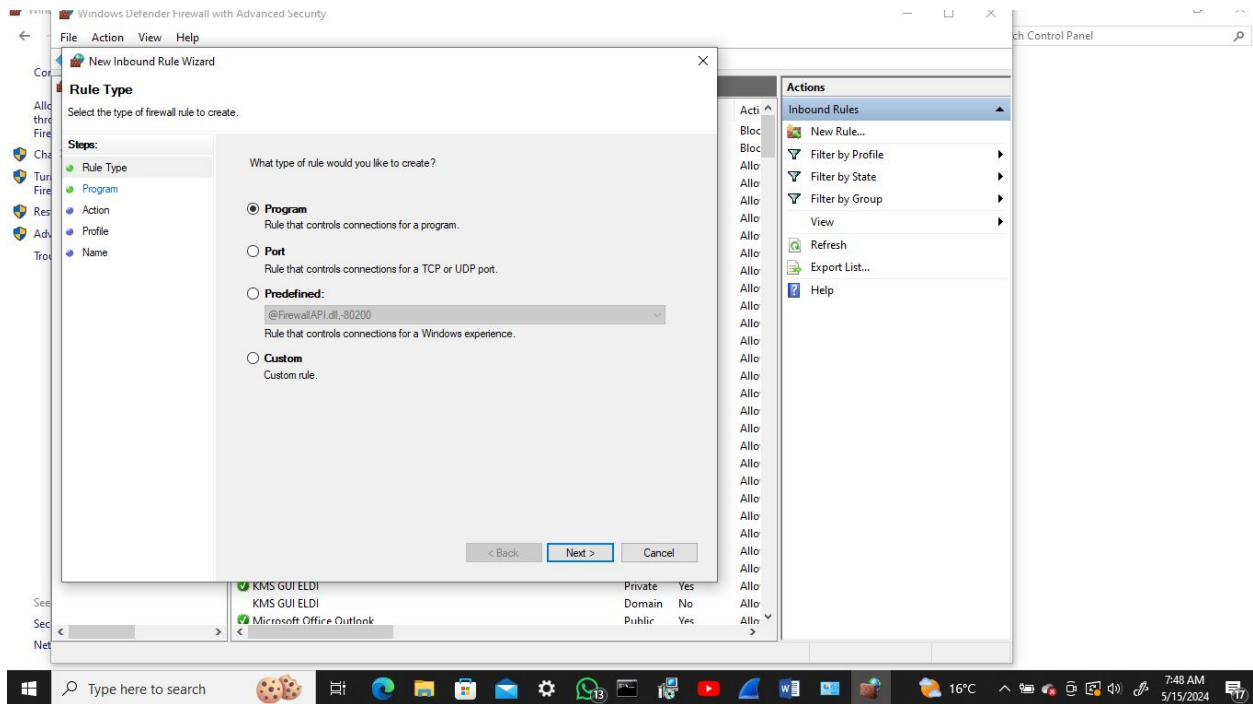
Allowing ICMP Traffic Through Firewall: Creating an inbound rule to let traffic through the firewall



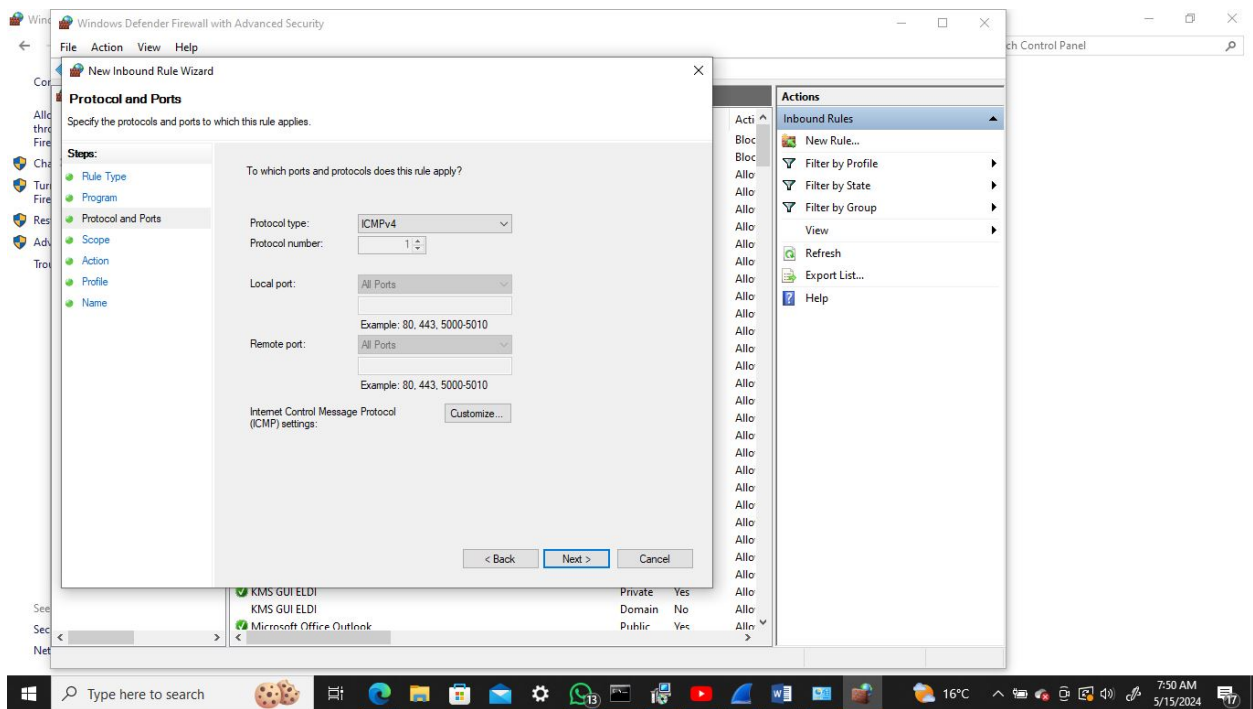
The first step was to navigate to windows defender firewall through the control panel.



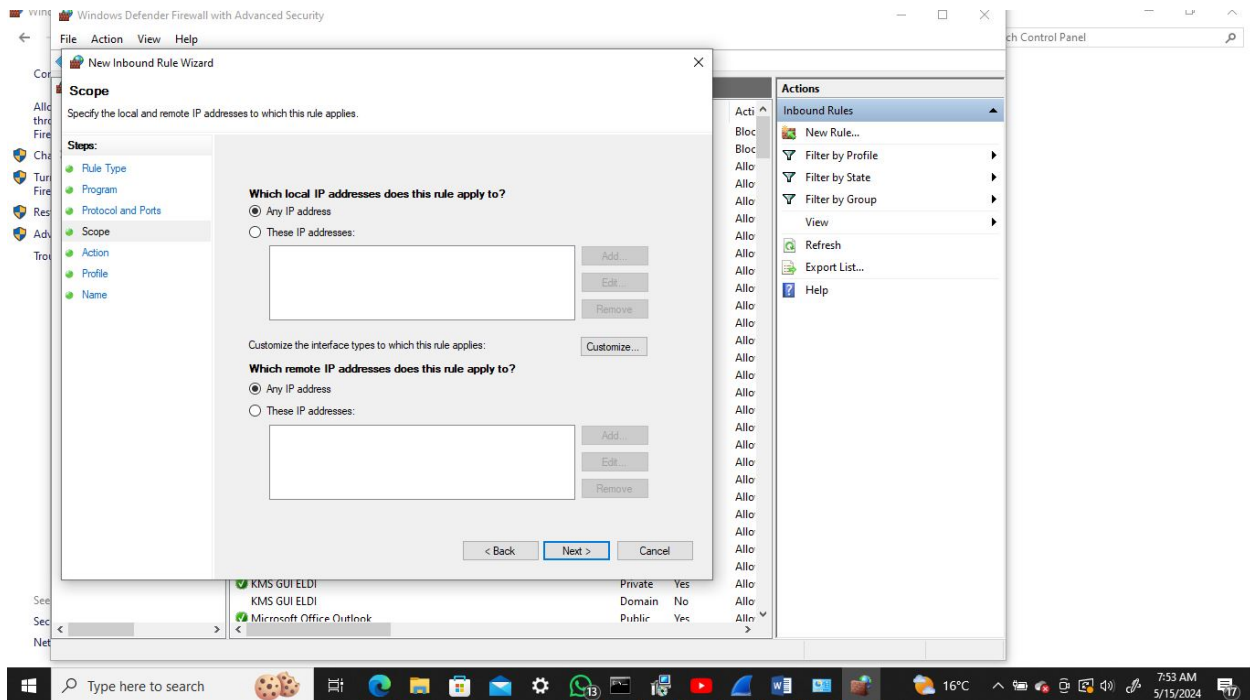
On the advanced security window, I created a new inbound rule



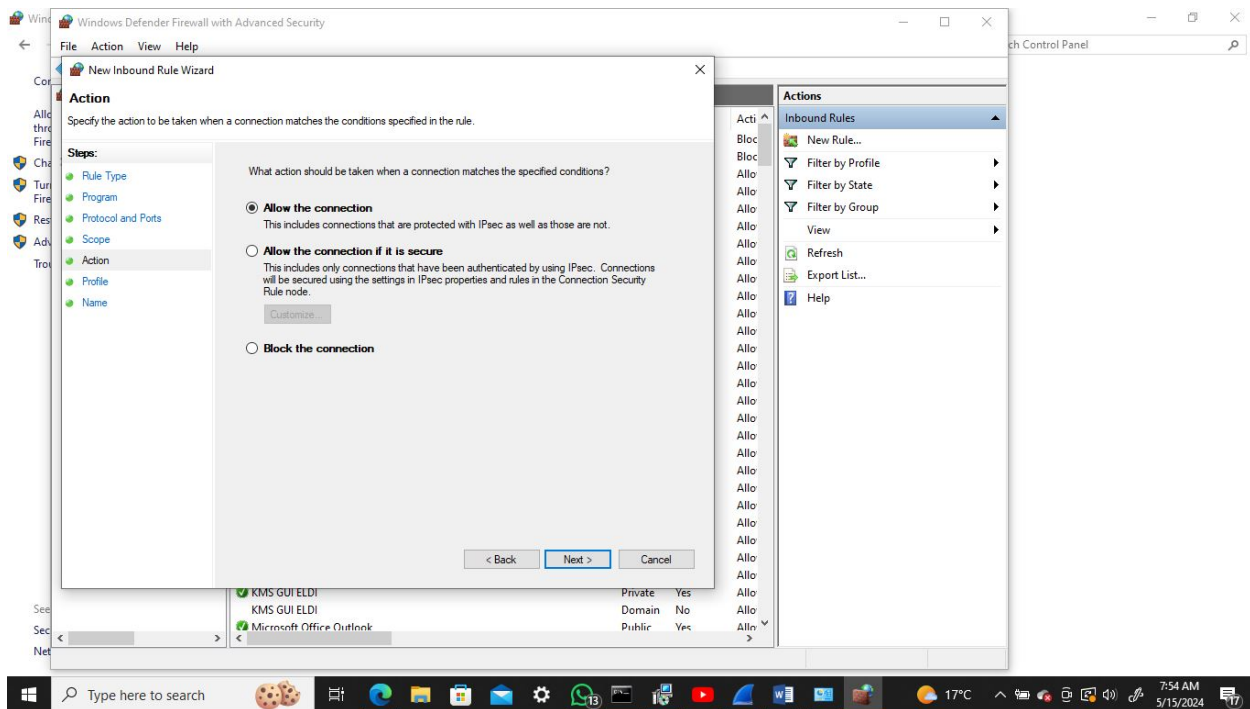
I created a custom rule using the New Inbound rule wizard.



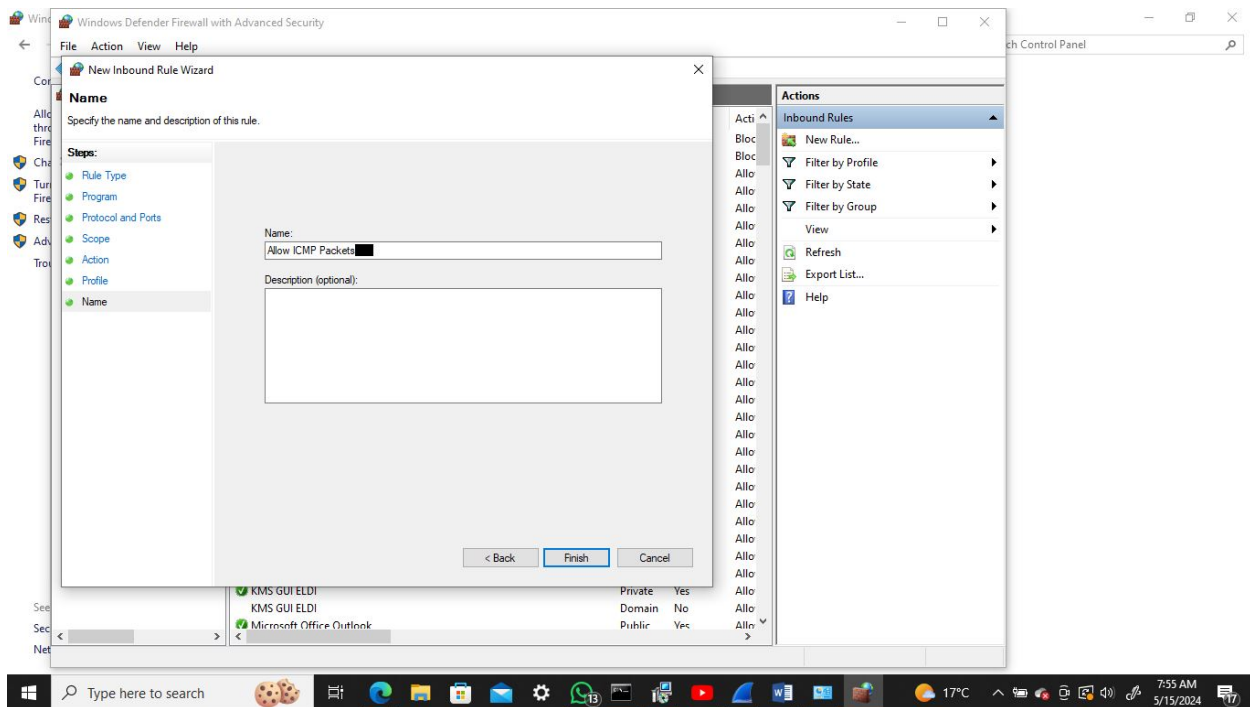
On the left hand side, I selected the protocols and ports. On the protocol type I input ICMP as it was the protocol I was creating a rule for.



Verified that all IP addresses for both local and remote were selected.

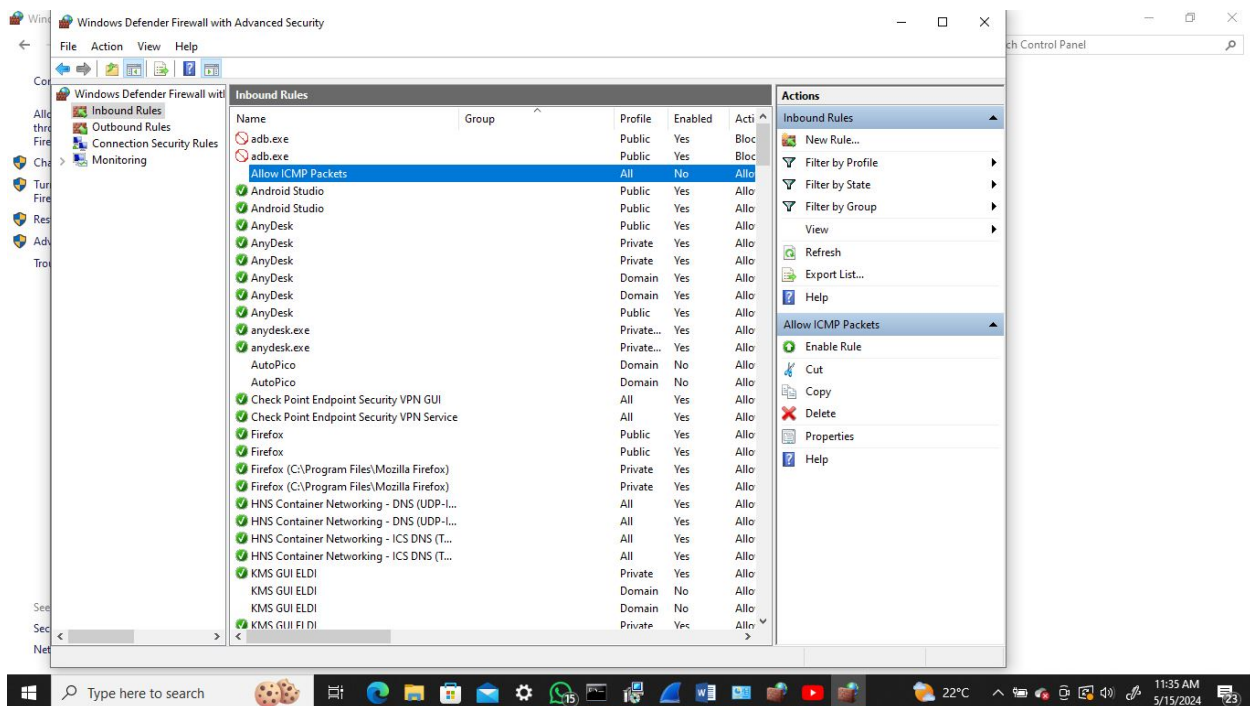


I then allowed the connection



This was the last step in the process where I named the newly created rule as, Allow ICMP Packets.

Disabling or Deleting the ICMP Rule



I navigated to the Inbound rule section window again in order to disable the ICMP Inbound rule I had created. I right clicked on the rule and I chose to disable. It might save someone time than deleting the rule and having to start over at a later date.

Conclusion

The aim of this assignment was to equip us with skills on use of tools such as Wireshark and other utilities such as ping. Ping helped me to determine the IP addresses of the urls and was critical for this assignment. Wireshark was also cardinal to the completion of this assignment as it offered important information. It started with determining the IP and MAC addresses and ended with creating a firewall inbound rule and then learning how to disable or delete one.